

Training-Free Action Correction for VLA Model Failures via Language Feedback

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Abstract—Vision-Language-Action (VLA) models demonstrate strong semantic understanding yet exhibit systematic failures during deployment. The conditions under which these failures occur, and whether they can be corrected without retraining, remain poorly understood. In this paper, we take steps toward addressing this gap. We present CorrectVLA, a framework that translates task-level natural language corrections into additive action magnitude adjustments without modifying policy weights. A human provides a single task-level correction, applied uniformly across all rollouts without per-episode intervention. In simulation, CorrectVLA recovers execution misalignment failures across both in-distribution and OOD tasks. In real-robot experiments on a UFactory xArm7 under environment shift, CorrectVLA restores near-perfect success where the base policy almost entirely breaks down, generalizing across object locations and identities. Through a taxonomy of failure modes on LIBERO-90, we find that execution misalignment failures, where the policy reaches the correct target but miscalibrates action magnitudes, represent the correctable subset, while other failure modes where semantic comprehension itself breaks down are not amenable to this approach. The approach succeeds when policies possess strategic correctness and fails when fundamental comprehension is absent, establishing a practical operational boundary for inference-time correction. Project Page: <https://correctvla.github.io>.

I. INTRODUCTION

Robotic foundation models have transformed manipulation through large-scale vision-language-action (VLA) architectures [1]–[5] that combine billion-parameter vision-language backbones with continuous motor control. Pretrained on diverse datasets, these models demonstrate remarkable semantic capabilities: recognizing objects in cluttered scenes, understanding natural language instructions, generalizing across tasks and embodiments. This semantic understanding enables VLAs to tackle diverse manipulation challenges, from tabletop rearrangement to complex multi-step assembly tasks.

Yet deployment to new environments reveals a persistent challenge. When test conditions diverge from training, new lighting, camera viewpoints, or table surfaces, VLAs may maintain semantic understanding but produce systematically biased action magnitudes, or fail in more fundamental ways where the policy misidentifies the target object or loses track of task goals entirely. The conditions under which these failures occur, and whether they can be corrected without retraining, remain poorly understood.

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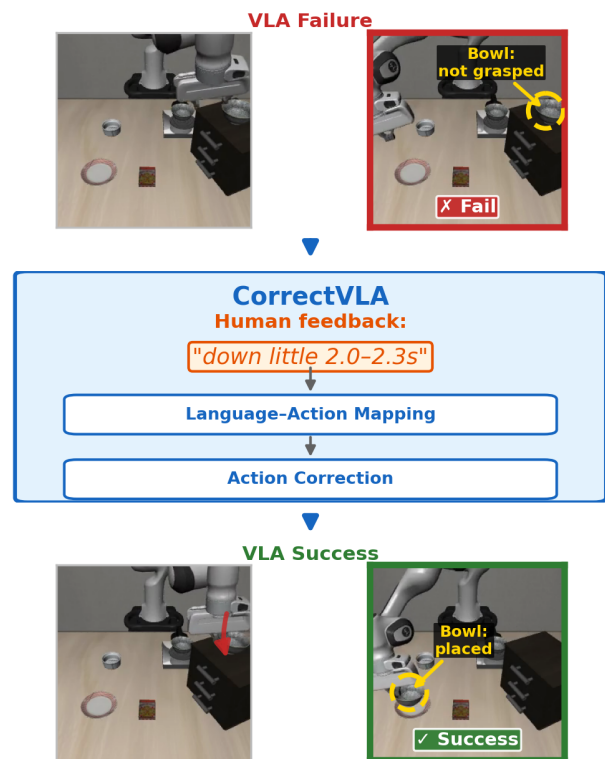


Fig. 1: **CorrectVLA Overview.** When a VLA policy fails to grasp the target object (top), a human provides natural language feedback (e.g., “down little 2.0–2.3s”). CorrectVLA grounds this into structured action corrections via language-action mapping, applied as additive biases to the frozen policy. Re-execution with corrected actions produces success (bottom).

Current approaches to deployment adaptation address this challenge through two complementary directions. Training-time methods collect new demonstrations and update model weights [2], [6], requiring infrastructure, GPU hours, and expertise. Test-time steering methods maintain frozen policies while using learned models or verification to select actions [7]–[9]. However, a gap remains: these approaches require either retraining infrastructure or cannot generate corrective behaviors for systematic magnitude errors not seen during training. As robot deployment scales beyond research labs to diverse real-world settings, an open question emerges: *can robots adapt to execution failures during deployment without retraining?*

In this paper, we take steps toward addressing this gap.

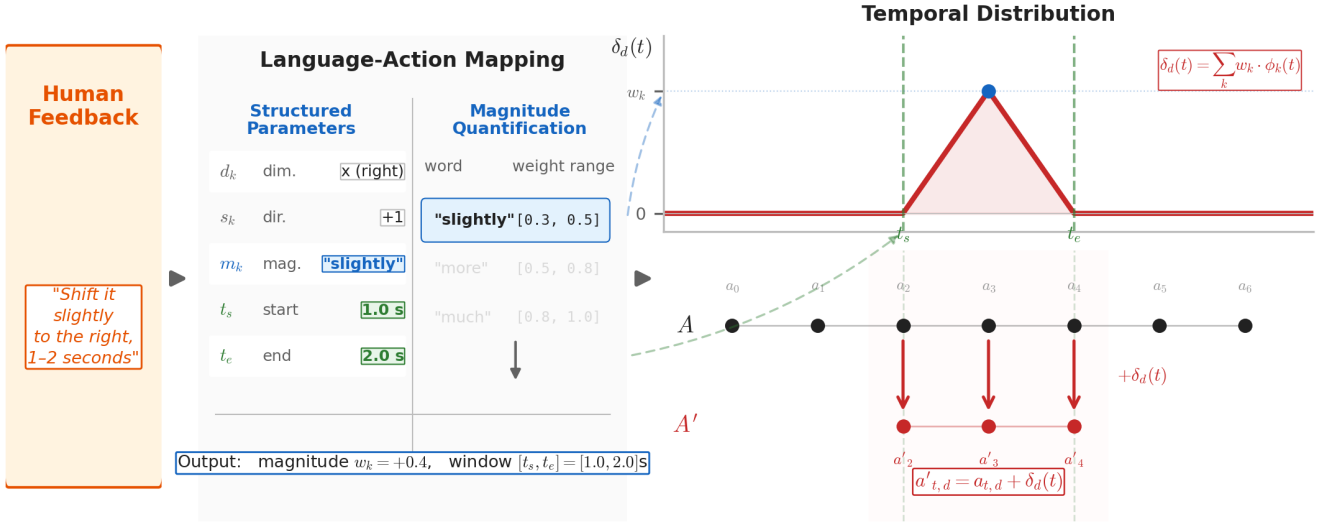


Fig. 2: **Language-to-Action Grounding Pipeline.** Human feedback (“shift it slightly to the right by 1–2 seconds”) is processed through two stages. **Left (Language-Action Mapping):** An LLM extracts structured parameters, dimension (d_k : x-axis), direction (s_k : +1 for right), and magnitude term (m_k : “slightly”), and quantifies the magnitude via word-to-weight mapping (“slightly” $\rightarrow$ 0.4 from range [0.3, 0.5]), producing output magnitude $w_k = 0.4$ and temporal window $[t_s, t_e] = [1.0, 2.0]$ s. **Right (Temporal Distribution):** A piecewise-linear envelope $\phi(t)$ distributes the correction smoothly across the window, ramping from 0 to peak at midpoint then back to 0. Red arrows show corrected actions $a^{\text{corrected}}$; black dots show original policy outputs a . The corrected trajectory A' deviates from A only within the specified window.

Through a taxonomy of failure modes on LIBERO-90, we find that not all failures are equal: execution misalignment failures, where the policy reaches the correct target but miscalibrates action magnitudes, represent a correctable subset, while failures rooted in semantic comprehension breakdown are not amenable to inference-time correction. This distinction motivates a targeted approach: rather than defaulting to retraining, we ask whether sparse human feedback can correct the failures that are actually correctable.

Our key insight separates two distinct problems: understanding task semantics versus calibrating execution magnitudes. When a robot overshoots a grasp, the semantic knowledge (which object to target, how to approach it, when to close the gripper) remains correct. Only the action magnitude for this deployment context requires adjustment. Human feedback naturally exploits this separation: rather than demonstrating the entire task again, people provide sparse corrections (“move left more,” “reach further”) that directly identify the calibration error (which action dimension drifted, in which direction, during which temporal window). This linguistic feedback enables targeted magnitude adjustment without questioning the policy’s strategic understanding, preserving its semantic capabilities while fixing execution drift.

We present CorrectVLA, a framework that translates natural language corrections into additive action magnitude adjustments without modifying VLA weights. Using LLM-based grounding, the system extracts structured parameters from linguistic feedback and constructs temporal corrections applied as per-dimension biases to the frozen policy’s

outputs. Execution failures exhibit temporal sparsity, errors concentrate at specific trajectory bottlenecks rather than pervading entire rollouts, and corrections applied at a small fraction of timesteps enable autonomous completion for the remainder. Each correction requires only seconds of human review versus hours for retraining, while preserving the VLA’s semantic generalization capabilities.

Contributions.

- A taxonomy of VLA deployment failures on LIBERO-90, identifying four consistent failure modes across OpenVLA-OFT [10] and $\pi 0.5$ [5] and establishing which are amenable to inference-time correction.
- CorrectVLA, a three-stage grounding mechanism mapping sparse linguistic feedback to dense per-timestep, per-dimension action adjustments, where sparse corrections enable autonomous completion without policy updates.
- Empirical validation on LIBERO demonstrating recovery of execution misalignment failures across both in-distribution and out-of-distribution tasks, with real-robot experiments achieving 95% success (19/20) after environment shift where the base policy drops to 10% (2/20).

II. RELATED WORK

A. Vision-Language-Action Models and Deployment Challenges

Generalist VLA models [1]–[4], [11] demonstrate strong cross-task generalization through large-scale pretraining on diverse robot datasets. OpenVLA [1] builds on Llama 2 7B

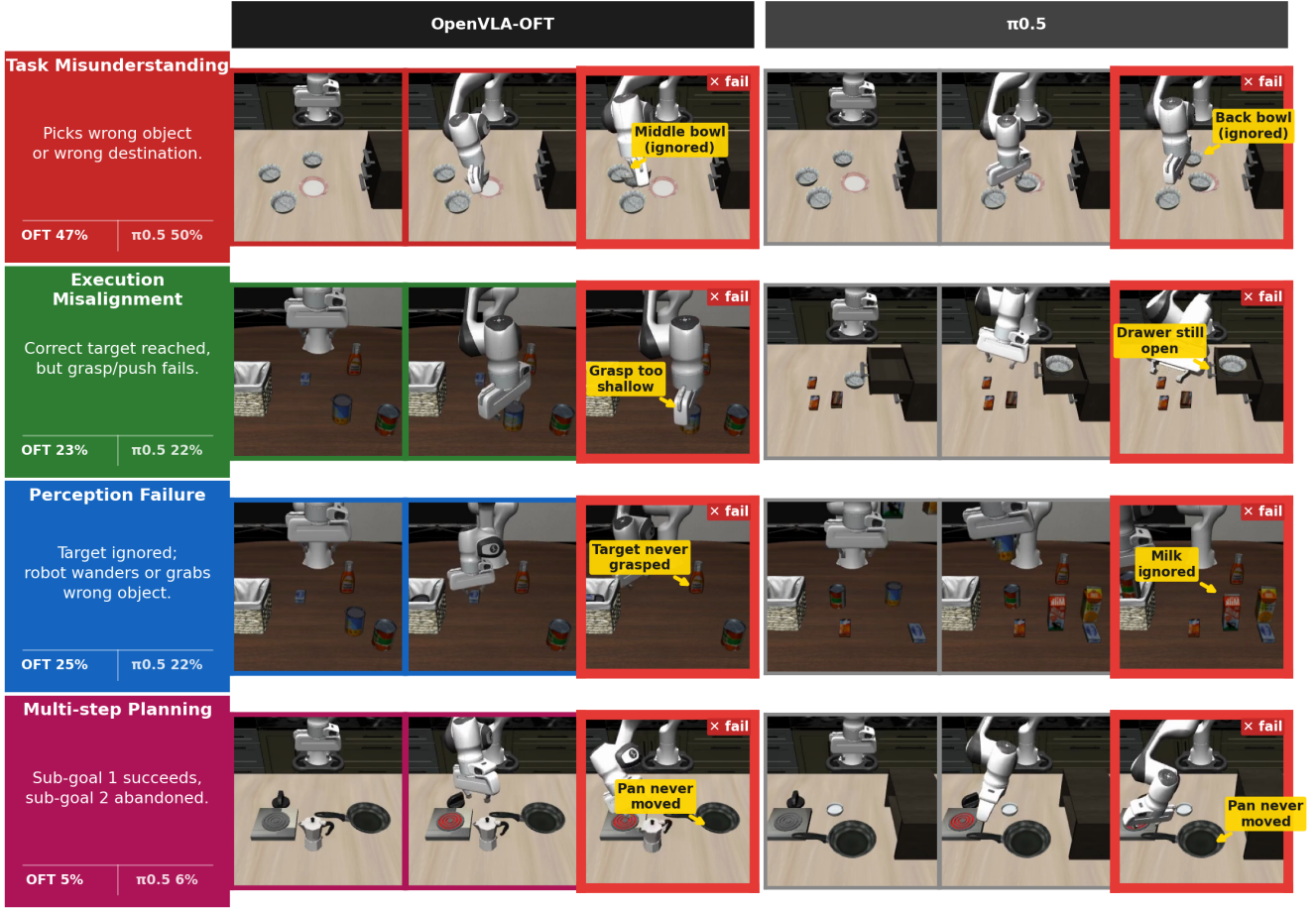


Fig. 3: **Failure Taxonomy of Generalist VLA Policies.** Four failure modes across 133 OOD tasks (73 OpenVLA-OFT, 60 $\pi 0.5$) on LIBERO-90, shown side by side for each model (both fine-tuned on the LIBERO dataset). *Task Misunderstanding* (red, $\sim 47\%/50\%$): the policy manipulates the wrong object due to misread spatial or relational language. *Execution Misalignment* (green, $\sim 23\%/22\%$): the policy reaches the correct target but miscalibrates action magnitude, grasp too shallow, push too weak. *Perception Failure* (blue, $\sim 25\%/22\%$): the target object is ignored entirely. *Multi-step Planning* (purple, $\sim 5\%/6\%$): the first sub-goal succeeds but the second is abandoned. Red borders indicate the critical failure frame. Only Execution Misalignment is amenable to inference-time correction.

with a ViT vision encoder, while RT-1/RT-2 [4], [11] generalize across 700+ tasks. Octo [3] provides a generalist policy trained on 800k trajectories across multiple embodiments. Recent Pi models [12] handle complex real-world tasks from espresso making to laundry folding.

Despite these capabilities, deployment to new environments exposes systematic vulnerabilities. Action magnitudes miscalibrate under lighting or viewpoint changes, fine-tuning requires hours of GPU training and new demonstrations, and deployment demands extensive infrastructure. More critically, the field lacks systematic understanding of *why* policies fail and *which* interventions are appropriate for a given failure. Our work addresses this gap by characterizing failure modes to identify when inference-time correction is viable, rather than defaulting to retraining.

B. Runtime Policy Correction and Steering

Recent work enables test-time adaptation without weight updates. ITPS [7] uses human physical corrections during diffusion sampling, learning task feature weights through online gradient descent, but requires continuous human guidance throughout execution. FOREWARN [9] integrates world models and VLM verifiers to score sampled trajectories, selecting optimal candidates, yet computational overhead from generating multiple samples limits real-time deployment. Mechanistic interpretability [13] modulates internal activations along semantic directions, but requires architecture-specific analysis and white-box access to model internals.

A fundamental limitation shared by these methods is that sampling-based approaches operate only within the learned output distribution, while verification systems provide no corrective mechanism for systematic magnitude errors. CorrectVLA takes a complementary approach: by directly apply-

ing additive corrections to action magnitudes, it can generate corrective behaviors outside the original distribution while keeping policy weights frozen.

C. Human-in-the-Loop Robot Learning

Human feedback approaches recognize that qualitative corrections outperform dense quantitative specifications [14], [15]. Language-guided trajectory improvement [16], [17] uses comparative feedback for preference learning, but demands multiple execution rounds and pairwise comparisons. Shared autonomy [18] maintains Bayesian beliefs over user goals for blended control, adapting during execution based on inferred intentions. ITPS [7] integrates physical corrections into diffusion generation, updating feature weights online through gradient-based learning.

These methods require either real-time supervision during execution or policy retraining with accumulated demonstrations. CorrectVLA instead uses sparse retrospective language corrections applied after observing a complete failure, requiring minimal human review per task without fine-tuning or continuous guidance.

III. PROBLEM FORMULATION

A. Problem Statement

Given a frozen VLA policy $\pi_\theta : \mathcal{O} \times \mathcal{L} \rightarrow \mathbb{R}^D$ and a failed trajectory $\tau_{\text{fail}} = \{(o_t, a_t)\}_{t=1}^T$, we seek a correction function $\delta : [T] \times [D] \rightarrow \mathbb{R}$ such that re-execution with corrected actions $a_{t,d}^{\text{corrected}} = a_{t,d} + \delta_{t,d}$ produces successful task completion. The correction must satisfy three properties: *sparsity* (specified by $K \ll T$ human feedback instances), *policy preservation* (weights θ remain frozen), and *deployment feasibility* (no new demonstrations or GPU training required). The central challenge is constructing the grounding function $\mathcal{G} : \mathcal{F}^K \rightarrow \mathbb{R}^{T \times D}$ that maps sparse natural language feedback to dense per-timestep, per-dimension action corrections.

B. Failure Modes of Generalist Policies

Analyzing 133 task-model evaluations (73 OpenVLA-OFT, 60 $\pi 0.5$) across LIBERO-90 out-of-distribution tasks reveals four distinct failure modes (Figure 3). *Task Misunderstanding* dominates for both models (OFT 47%, $\pi 0.5$ 50%): the policy misreads spatial or relational language and manipulates the wrong object. *Perception Failure* accounts for 25%/22% of tasks, where the target object is ignored entirely. *Execution Misalignment* (23%/22%) occurs when the policy reaches the correct target but fails to complete the action, grasp too shallow, push too weak. *Multi-step Planning* failure is rare (5%/6%): the first sub-goal succeeds but the second is abandoned.

Critically, these modes differ in their amenability to inference-time correction. Task Misunderstanding and Perception Failure reflect a breakdown in semantic comprehension, the policy has fundamentally misidentified the task or target, and magnitude adjustment alone cannot compensate. Execution Misalignment, by contrast, preserves strategic correctness while miscalibrating action magnitudes, making it the primary candidate for the correction approach developed in Section IV.

C. Assumptions

Our framework operates under three assumptions.

Assumption 1 (Strategic Correctness). The policy π_θ understands the task and reaches near-successful execution; failures arise solely from action magnitude miscalibration.

Assumption 2 (Sparse Correctability). A small number of per-dimension magnitude adjustments $\delta_{t,d}$ are sufficient to convert a near-miss into task success.

Assumption 3 (Human Observability). A human reviewer can identify the critical error region and provide sparse corrections $\mathcal{F} = \{f_1, \dots, f_K\}$ after observing the complete rollout.

IV. METHODS

CorrectVLA constructs corrections through three phases: extracting structured parameters from natural language (Phase A), mapping linguistic terms to numeric magnitudes (Phase B), and distributing corrections temporally (Phase C). Feedback is provided once at the task level and reused across all rollouts without per-episode intervention.

A. Phase A: Structured Parameter Extraction

Given natural language feedback, we use an LLM to extract four structured parameters. For example, “move left more in 1–2 seconds” is parsed into an action dimension, direction, magnitude term, and temporal window. This produces a structured correction $c_k = (d_k, s_k, m_k, t_{\text{start},k}, t_{\text{end},k})$, where $d_k \in \{x, y, z, \text{roll}, \text{pitch}, \text{yaw}, \text{grasper}\}$ is the action dimension, $s_k \in \{+1, -1\}$ indicates direction, $m_k \in \{\text{slightly}, \text{more}, \text{much}\}$ is the magnitude term, and $[t_{\text{start},k}, t_{\text{end},k}]$ specifies the temporal window.

B. Phase B: Magnitude Quantification

The magnitude term from Phase A is mapped to a numeric value using empirically validated ranges: “slightly” maps to $[0.3, 0.5]$, “more” to $[0.5, 0.8]$, and “much” to $[0.8, 1.0]$. The LLM selects a value within the appropriate range and applies the direction from Phase A to produce a signed magnitude $w_k \in [-1.0, 1.0]$. For example, “more” in the negative direction produces $w_k = -0.65$. These ranges were validated empirically on a held-out set of correction examples prior to evaluation.

C. Phase C: Temporal Distribution

To ensure smooth integration with the policy’s temporal dynamics, we construct a piecewise-linear envelope $\phi_k(t)$ that distributes the correction across the specified window, ramping from 0 to peak magnitude at the midpoint then back to 0:

$$\phi_k(t) = \begin{cases} 0 & t < t_{\text{start},k} \\ \frac{2(t - t_{\text{start},k})}{t_{\text{end},k} - t_{\text{start},k}} & t_{\text{start},k} \leq t \leq t_{\text{mid},k} \\ \frac{2(t_{\text{end},k} - t)}{2(t_{\text{end},k} - t_{\text{mid},k})} & t_{\text{mid},k} < t \leq t_{\text{end},k} \\ 0 & t > t_{\text{end},k} \end{cases} \quad (1)$$

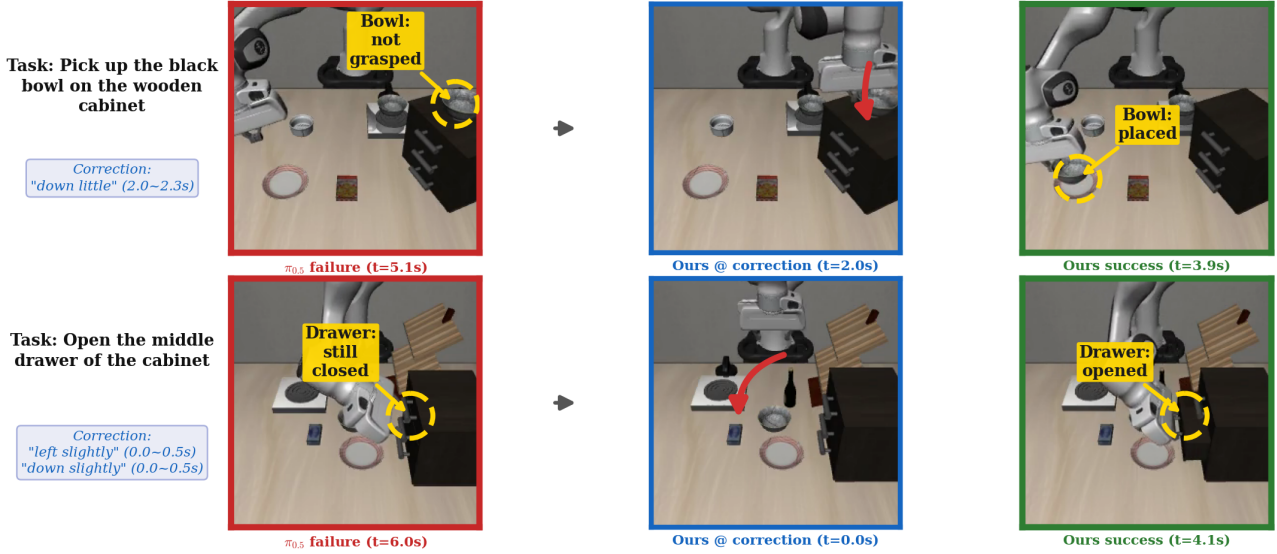


Fig. 4: **Correction Examples.** $\pi_{0.5}$ failures (left, red) are recovered by sparse task-level corrections (center, blue), producing successful execution (right, green). **Top** (*Pick up the black bowl on the wooden cabinet*): a single correction (“down little, 2.0–2.3s”) redirects the gripper to recover a missed bowl grasp. **Bottom** (*Open the middle drawer of the cabinet*): two corrections (“left slightly, 0.0–0.5s” and “down slightly, 0.0–0.5s”) steer the approach to recover a drawer that was never opened. In both cases, the policy’s high-level structure remains intact; only the action magnitude at the critical bottleneck requires adjustment.

where $t_{\text{mid},k} = (t_{\text{start},k} + t_{\text{end},k})/2$. This localization concentrates corrections at critical trajectory moments while maintaining zero influence outside the specified window.

D. Complete Grounding Function

For K feedback instances $\{f_1, \dots, f_K\}$, the grounding function aggregates corrections across dimensions:

$$\mathcal{G}(f_1, \dots, f_K) = \delta, \quad \delta_d(t) = \sum_{k: d_k=d} w_k \cdot \phi_k(t) \quad (2)$$

The corrected action at timestep t and dimension d is then:

$$a_{t,d}^{\text{corrected}} = a_{t,d} + \delta_d(t) \quad (3)$$

By construction, corrections are sparse in feedback space, localized in time, and superpose additively across dimensions. Policy weights remain frozen throughout, and task-level feedback generalizes across all rollouts without per-episode human intervention.

V. EXPERIMENTAL SETUP

A. Benchmark and Models

We evaluate on LIBERO [19] (MuJoCo, Franka Panda 7-DoF) for simulation and a UFactory xArm7 with Robotiq 2F-85 gripper and three RealSense cameras for real-robot experiments. For the failure taxonomy, we analyze two base policies: OpenVLA-OFT [10], a 7B-parameter VLA fine-tuned on 40 LIBERO tasks, and $\pi_{0.5}$ [5], fine-tuned on LIBERO (lerobot/pi05-libero-finetuned) for simulation and on the DROID [20] for real-robot experiments.

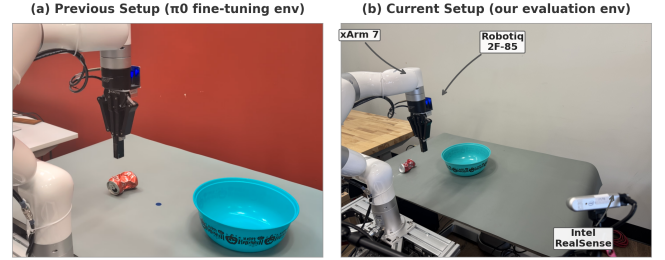


Fig. 5: **Real-Robot Experimental Setup.** (a) **Previous setup** ($\pi_{0.5}$ fine-tuning environment): the policy was fine-tuned with a fixed camera viewpoint and table configuration. (b) **Current setup** (our evaluation environment): the robot base is relocated to a new position, introducing distribution shift in camera viewpoint and workspace geometry. The xArm 7 with Robotiq 2F-85 gripper and Intel RealSense camera are used for all real-robot experiments. This environment shift, same objects, different robot pose, is the primary source of execution misalignment evaluated in Section VI.

B. Baselines

We compare three conditions.

Base Policy: the frozen policy with no correction applied, serving as the lower bound.

LLM Baseline: a VLM autonomously watches the failure video and generates corrections using its own prior reasoning, inferring per-axis displacement and applied bias, then adjusting magnitude and timing based on observed overshoot or undershoot, without any human input; the LLM Baseline

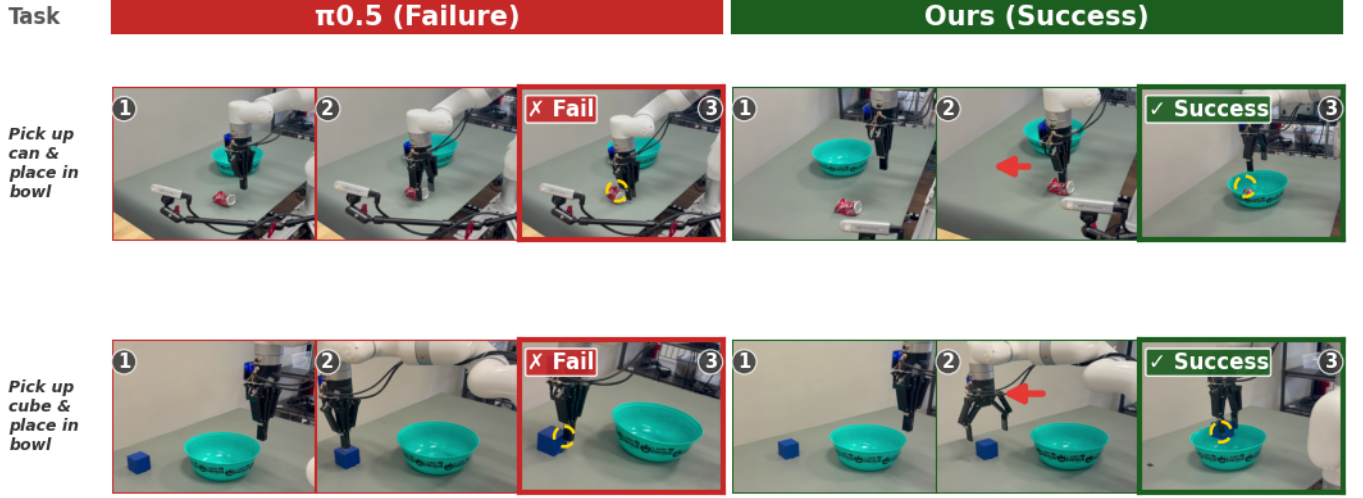


Fig. 6: **Real-Robot Results.** After shifting the robot base, $\pi 0.5$ fails on both pick-and-place tasks due to execution misalignment (left, red): the gripper misses the target on both *pick up can and place in bowl* and *pick up cube and place in bowl*. CorrectVLA recovers success on both tasks using a single task-level correction (right, green), without per-episode intervention or retraining.

TABLE I: Simulation results across in-distribution and out-of-distribution evaluations. CorrectVLA is applied only to tasks exhibiting execution misalignment failures.

Setting	Metric	$\pi 0.5$ -LIBERO	VLM Base	Ours
In-Dist.	Success rate	1861 (93.0%)	—	1926 (96.3%)
	Recovery rate	0%	0%	65/139 (46.8%)
OOD	Success rate	122 (27.1%)	—	155 (34.4%)
	Recovery rate	0%	0%	33/328 (10.1%)

uses GPT-5 mini.

CorrectVLA (Ours): a human provides task-level natural language corrections once per task, reused across all rollouts without per-episode intervention. Human corrections are formatted as JSON via Claude Sonnet 4.5.

C. Evaluation Protocol

Simulation. We run 50 trials per task on 40 in-distribution tasks (2,000 rollouts) and 5 trials per task on all 90 LIBERO-90 tasks (450 rollouts) for out-of-distribution evaluation. Corrections are applied only to tasks where the base policy exhibits execution misalignment failures, consistent with our taxonomy.

Real Robot. All experiments are based on the RIO framework [21]. We evaluate pick-and-place tasks under three generalization conditions: (1) same object, changed environment, (2) same object, different location, and (3) different object, different location. Fine-tuning data was collected via Spacemouse and GELLO interfaces; Spacemouse

TABLE II: Real-robot results (pick-and-place, can) after robot base shift. CorrectVLA recovers across all three generalization conditions with a single task-level correction.

OOD Variation		$\pi 0.5$ -DROID [†]	VLM Base	Ours
Location	Object			
Same	Same	1/10 (10%)	—	10/10 (100%)
Diff	Same	0/5 (0%)	—	5/5 (100%)
Diff	Diff	1/5 (20%)	—	4/5 (80%)
Total (20 trials)		2/20 (10%)	—	19/20 (95%)

[†] $\pi 0.5$ -DROID achieves 95% (19/20) on pick-and-place in the original environment before base shift.

yielded cleaner demonstrations for tasks where rotation is minimal. All experiments run on an NVIDIA RTX 4090, AMD Ryzen 7 5700X, 64 GB RAM. The $\pi 0.5$ fine-tuned model achieves 95% success over 20 trials in the original setting. We then shift the robot base to a new position and test generalization across all three conditions.

D. Action Space Adaptation

Corrections are specified in Cartesian space to match human intuition (“move left,” “move up”). For the real-robot xArm7, which operates in joint velocity control, we convert Cartesian corrections to joint velocities via the Jacobian pseudoinverse:

$$\dot{\mathbf{q}} = \mathbf{J}^\dagger(\mathbf{q}) \dot{\mathbf{x}} \quad (4)$$

where $\mathbf{J}(\mathbf{q})$ is the manipulator Jacobian at the current configuration $\mathbf{q}$ and $\dot{\mathbf{x}}$ is the desired Cartesian correction. This yields the minimum-norm joint velocity achieving the target end-effector motion.

VI. RESULTS

A. Simulation Results

1) *Failure Taxonomy*: Analysis of OOD failures on LIBERO-90 reveals four consistent modes (Figure 3): Task Misunderstanding ($\sim 50\%$), Perception Failure ($\sim 22\%$), Execution Misalignment ($\sim 22\%$), and Multi-step Planning ($\sim 6\%$) for $\pi 0.5$. Only Execution Misalignment maintains strategic correctness and is thus amenable to inference-time correction; the remaining modes require training-level interventions.

2) *In-Distribution Recovery*: $\pi 0.5$ achieves 93.0% in-distribution success (1861/2000) across all four suites: LIBERO-Spatial 86.8% (434/500), LIBERO-Object 99.8% (499/500), LIBERO-Goal 89.6% (448/500), and LIBERO-10 96.0% (480/500), leaving 139 failures. CorrectVLA recovers 65 of these through task-level corrections applied uniformly across rollouts, achieving a recovery rate of 46.8% (65/139) and raising overall success to 96.3% (1926/2000). The LLM baseline recovers nothing, as autonomous correction from visual feedback alone proves insufficient for reliable magnitude estimation.

3) *Out-of-Distribution Recovery*: On all 90 LIBERO-90 tasks (450 rollouts), $\pi 0.5$ achieves 27.1% success (122/450), leaving 328 failures. CorrectVLA recovers 33 execution misalignment failures, achieving a recovery rate of 10.1% (33/328) and raising overall success to 34.4% (155/450). For example, all 4 failures on *open_the_top_drawer* and all 5 on *put_the_black_bowl_on_the_plate* are recovered with a single task-level correction each. Task misunderstanding failures are not recovered, consistent with our taxonomy. The LLM baseline again recovers nothing.

B. Real-Robot Results

We evaluate pick-and-place under significant distribution shift: after relocating the robot base, $\pi 0.5$ drops from 95% (19/20) in the original setting to 10% (2/20). CorrectVLA recovers performance across all three generalization conditions using a single task-level correction (Table II): 100% on same location (10/10), 100% on different location (5/5), and 80% on different object and location (4/5), achieving 95% overall success (19/20). The LLM baseline recovers nothing across all conditions, consistent with simulation findings.

VII. DISCUSSION

Failure taxonomy and intervention boundaries Figure 3 reveals consistent failure patterns across both models: Task Misunderstanding (47%/50%) and Perception Failure (25%/22%) together account for over 70% of failures in both OpenVLA-OFT and $\pi 0.5$. These modes share a common cause, VLA models develop strong scene priors during training, making them resistant to steering when the wrong object is targeted or ignored entirely. Magnitude adjustment alone cannot compensate for absent comprehension; such failures require architectural or training-level interventions. By contrast, Execution Misalignment (23%/22%) preserves correct task understanding while miscalibrating action magnitudes, making it the primary candidate for inference-time

correction. Importantly, the consistency of this distribution across two architecturally distinct models suggests these failure modes reflect general properties of generalist VLA policies under distribution shift, rather than model-specific artifacts.

Sparse corrections propagate through policy dynamics Figure 4 illustrates a key finding: a small number of targeted action adjustments at critical trajectory moments are sufficient to convert failure into success. For *pick up bowl on cabinet*, a single correction (“down little, 2.0–2.3s”) redirects the gripper to achieve a successful grasp; for *open middle drawer*, two corrections at $t=0s$ steer the approach correctly. In both cases, the policy’s high-level structure, approach, grasp, transport, remains intact; only the magnitude at the critical bottleneck requires adjustment. This supports a compositional view of VLA execution: task semantics and motor execution are separable, and correcting one does not disrupt the other.

Real-robot generalization Figure 6 demonstrates that task-level corrections transfer reliably to real-robot deployment under significant distribution shift. After relocating the robot base, $\pi 0.5$ alone drops from 95% to 10% success on both pick-and-place tasks due to misaligned grasps. A single task-level correction restores success across same-location, different-location, and different-object conditions, confirming two properties of CorrectVLA: corrections generalize across rollout variations within a task, and execution misalignment under environment shift is addressable without per-episode intervention or retraining.

VIII. LIMITATIONS AND FUTURE WORK

Limitations. Our framework has three notable limitations. First, the language-to-magnitude mapping samples w_k from predefined ranges (“slightly” $\rightarrow [0.3, 0.5]$, “more” $\rightarrow [0.5, 0.8]$, “much” $\rightarrow [0.8, 1.0]$), introducing variability between intended and applied correction magnitudes; convergence may require multiple rounds. Second, task-level corrections assume a single representative failure captures the dominant misalignment pattern for that task, if failure modes vary significantly across episodes, a single correction may not generalize to all rollouts. Third, our failure taxonomy is derived from simulation; real-world failure distributions may differ in ways that affect both the taxonomy and the applicability of inference-time correction.

Future Directions. Several directions follow naturally from these limitations. *Learning from experience*: rather than relying on predefined magnitude ranges, learning from accumulated correction failures could enable adaptive magnitude estimation, reducing the number of correction rounds needed [22]. *Broader failure modes*: object-centric architectures and domain randomization may address Task Misunderstanding and Perception Failure, while explicit spatial reasoning could reduce planning failures. *Retrieval validation*: extending the taxonomy to physical deployments and integrating automatic failure mode detection with retrieval-based correction [23], [24] would strengthen the operational boundary established here.

IX. CONCLUSION

We investigate when and how VLA deployment failures can be corrected at inference time. Through a taxonomy of failures across 133 task-model evaluations on LIBERO-90, we identify four consistent modes across OpenVLA-OFT and $\pi 0.5$: Task Misunderstanding ($\sim 50\%$), Perception Failure ($\sim 23\%$), Execution Misalignment ($\sim 23\%$), and Multi-step Planning ($\sim 6\%$). These modes differ fundamentally in their amenability to inference-time correction: Execution Misalignment, where strategic correctness is preserved but action magnitudes are miscalibrated, represents the correctable subset, while failures rooted in semantic comprehension breakdown require training-level interventions.

We present CorrectVLA, a framework that translates task-level natural language corrections into additive action adjustments without modifying policy weights. In simulation, CorrectVLA recovers execution misalignment failures across both in-distribution and OOD tasks using a single correction per task. In real-robot experiments under significant distribution shift, task-level corrections generalize across same-location, different-location, and different-object conditions, achieving 95% success where the base policy drops to 10%.

Together, these findings suggest a practical principle for robot deployment: rather than defaulting to retraining, diagnose the active failure mode and apply the minimal sufficient intervention. The approach succeeds when policies possess strategic correctness and fails when fundamental comprehension is absent, a boundary we hope will guide future work on deployment-time adaptation.

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